

10.4 HOW FIELDS AND OPERATORS TRANSFORM

This note collects the transformation laws used from the first weeks of the course, with the same conventions as §10.3: $U(1 + \delta\omega) = I + \frac{i}{2}\delta\omega_{\mu\nu}M^{\mu\nu}$.

SOURCE AND TARGET MANIFOLDS

- **Source manifold:** the spacetime on which the coordinates x^μ are defined, here Minkowski space $\mathbb{R}^{1,3}$. Fields are functions on it.
- **Target manifold:** the space of values of a field. For a real scalar field it is $\mathbb{R}$; for a complex scalar field it is $\mathbb{C} \cong \mathbb{R}^2$. If the modulus $|\phi|$ is frozen (for example at low energies in a phase where the $U(1)$ symmetry is spontaneously broken), only the phase remains and the target is the circle S^1 .
- **Scalar field:** a map ϕ : source manifold $\rightarrow$ target manifold whose value at a spacetime point does not depend on the observer.

ACTIVE AND PASSIVE TRANSFORMATIONS

- **Active:** the field configuration is moved (rotated, boosted); the coordinate system is kept fixed.
- **Passive:** the observer's coordinates are relabeled, $x \rightarrow \bar{x} = \Lambda x$; the physical field is untouched.

Passive. Let Alice use coordinates x^μ and Bob $\bar{x}^\mu = \Lambda^\mu{}_\nu x^\nu$. For a scalar, both assign the same value to the same point P :

$$\bar{\phi}(\bar{x}) = \phi(x) \implies \bar{\phi}(\bar{x}) = \phi(\Lambda^{-1}\bar{x}), \quad \text{i.e.} \quad \bar{\phi}(x) = \phi(\Lambda^{-1}x).$$

Active. In QFT, Lorentz transformations act on the Hilbert space through unitary operators $U(\Lambda)$, and a scalar field operator obeys

$$U(\Lambda)^{-1} \phi(x) U(\Lambda) = \phi(\Lambda^{-1}x).$$

Taking expectation values, $\langle \psi | U^{-1} \phi(x) U | \psi \rangle = \langle \psi | \phi(\Lambda^{-1}x) | \psi \rangle$: in the transformed state $U(\Lambda)|\psi\rangle$, the field at x has the value it had at $\Lambda^{-1}x$ in $|\psi\rangle$. The configuration has been moved by Λ .

The two agree. Writing $\phi'(x) \equiv U(\Lambda)^{-1} \phi(x) U(\Lambda)$, the active and passive results are the same function,

$$\phi'(x) = \bar{\phi}(x) = \phi(\Lambda^{-1}x),$$

so the two views make identical predictions. Moving the system by Λ is equivalent to relabeling the coordinates by Λ .

880 **Example.** A temperature field $T(\mathbf{x})$ on $\mathbb{R}^3$ has target $\mathbb{R}$. Rotating the
 881 system by R gives $T'(\mathbf{x}) = T(R^{-1}\mathbf{x})$: the temperature now at $\mathbf{x}$ is the one
 882 that used to be at $R^{-1}\mathbf{x}$.

883 FOUR-VECTOR OPERATORS

884 Operators that carry a Lorentz index but no spacetime argument
 885 transform with Λ itself. The basic examples are the energy-momentum
 886 operator and the Lorentz generators:

$$U(\Lambda)^{-1}P^\mu U(\Lambda) = \Lambda^\mu{}_\nu P^\nu, \quad U(\Lambda)^{-1}M^{\mu\nu}U(\Lambda) = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma M^{\rho\sigma}.$$

887 Infinitesimally this is the commutator

$$[P^\mu, M^{\rho\sigma}] = i(g^{\mu\sigma}P^\rho - g^{\mu\rho}P^\sigma),$$

888 which, with $U(1 + \delta\omega) = I + \frac{i}{2}\delta\omega_{\rho\sigma}M^{\rho\sigma}$, gives $U^{-1}P^\mu U = P^\mu + \delta\omega^\mu{}_\nu P^\nu$. Unlike
 889 $\phi(x)$, P^μ is not a function of spacetime: its *components* mix under Λ ; there
 890 is no argument to transform.

891 WHY Λ^{-1} FOR THE ARGUMENT BUT Λ FOR 892 COMPONENTS?

- 893 • **Arguments:** the field at x after the transformation is the field that
 894 used to be at the point that was carried to x , namely $\Lambda^{-1}x$.
- 895 • **Components:** indices label directions, and directions are rotated or
 896 boosted along with the system, so they transform with Λ .
- 897 • **Both at once:** a vector field has an argument and a component
 898 index, $U(\Lambda)^{-1}V^\mu(x)U(\Lambda) = \Lambda^\mu{}_\nu V^\nu(\Lambda^{-1}x)$. The general case, with a
 899 spin part $S^{\mu\nu}$ for each representation, is in §10.3.

Object	$U(\Lambda)^{-1}(\cdot)U(\Lambda)$	What transforms
Scalar field $\phi(x)$	$\phi(\Lambda^{-1}x)$	the argument
Four-vector operator P^μ	$\Lambda^\mu{}_\nu P^\nu$	the components
Vector field $V^\mu(x)$	$\Lambda^\mu{}_\nu V^\nu(\Lambda^{-1}x)$	both

TRANSFORMATION LAWS UNDER $U(\Lambda)$

900 CONSISTENCY WITH THE GROUP COMPOSITION LAW

901 The rules must respect $U(\Lambda_1)U(\Lambda_2) = U(\Lambda_1\Lambda_2)$. Apply the two factors one
 902 after the other:

$$U(\Lambda_1\Lambda_2)^{-1}\phi(x)U(\Lambda_1\Lambda_2) = U(\Lambda_2)^{-1}\phi(\Lambda_1^{-1}x)U(\Lambda_2) = \phi(\Lambda_2^{-1}\Lambda_1^{-1}x) = \phi((\Lambda_1\Lambda_2)^{-1}x),$$

$$U(\Lambda_1\Lambda_2)^{-1}P^\mu U(\Lambda_1\Lambda_2) = U(\Lambda_2)^{-1}\Lambda_1^\mu{}_\nu P^\nu U(\Lambda_2) = \Lambda_1^\mu{}_\nu \Lambda_2^\nu{}_\rho P^\rho = (\Lambda_1\Lambda_2)^\mu{}_\rho P^\rho.$$

903 Both close correctly. Had we written $\phi(\Lambda x)$ for the scalar, the first line
 904 would give $\phi(\Lambda_2\Lambda_1 x)$, the wrong order. That is why the inverse appears in
 905 the argument.